

Closing the Variance Gap: Bochner PINN for Physics-Informed Learning on Stationary Random Fields

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Abstract

Standard PINNs trained on stochastic PDEs converge to the conditional mean of the solution law; MSE training provides no distributional signal. We introduce *Bochner PINN*, augmenting the standard loss with log-spatial PSD, log-temporal PSD, and variance ratio terms evaluated against a finite-difference reference on a randomised spectral sub-grid. For stationary *Gaussian* random fields with separable covariance, combining Bochner’s theorem with the Gelbrich–Givens–Shortt formula and a kernel-smoothed periodogram estimator yields a single-realisation W_2 bound $W_2 \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta_{\text{sr}}} + C_3\sqrt{\delta}$ with $\beta_{\text{sr}} = s_x/(2(2s_x + d))$ and explicit constants. For *Lévy*-driven fields the same bound holds for the Gaussian projection of the law (second-order structure). We validate on Lévy-driven SPDEs: variance ratio rises from 0.19 to 0.83 on a hierarchical Variance-Gamma SPDE, the same loss closes the gap on four of five FNO benchmarks.

Keywords: physics-informed neural networks, stochastic partial differential equations, stationary random fields, spectral methods, distributional convergence, Lévy processes

1. Introduction

1.1. Motivation

Many physical and financial systems are governed by stochastic PDEs whose solutions are random fields rather than deterministic functions. Accurate surrogate models for such systems must reproduce the *probability law* of the solution, not merely its mean, since downstream quantities such as tail risk, variance, and higher moments depend critically on the full distribution. Physics-informed neural networks (PINNs) are natural candidates for efficient surrogate modelling: their inference cost is $O(|\theta|)$, independent of spatial resolution and dimension once trained, making them attractive for real-time applications where classical solvers are prohibitively expensive. The problem addressed here is that standard PINN training systematically fails to recover the law: the MSE loss contains no distributional signal, so the network converges to the conditional mean of the random field and loses variance, tail structure, and spectral content.

The failure is especially acute for SPDEs driven by Lévy processes, which are the primary validation setting of this paper. Lévy-driven dynamics arise across financial risk management, statistical mechanics, and plasma physics (non-local heat transport in Tokamak reactors), and they capture jump discontinuities and heavy-tailed distributions essential for extreme climatic events and biological foraging under the Lévy Flight Foraging Hypothesis. The non-local fractional Laplacian operator compounds the difficulty: FD discretisations produce dense matrices ($O(N^{2d})$ storage, prohibitive at $N = 32$, $d = 4$) and MC methods converge at $O(n^{-1/2})$, rendering both impractical for real-time use. The distributional failure of vanilla PINNs therefore closes off the one computationally efficient path.

1.2. The Problem

The core difficulty is that standard PINN losses constrain neither the probability law of the random field nor its frequency content across realisations. The NTK eigenvalue spectrum governs which frequency components the network learns; low-frequency eigenvalues dominate standard training, so the network learns the general shape of the solution and discards the high-frequency structure that encodes variance and tail behaviour. This spectral-bias account is correct for deterministic PDEs but understates the stochastic problem: the relevant learning target is the *probability law* of the random field, and MSE on collocation points provides no signal about it whatsoever.

For the Lévy-driven SPDEs studied here, the problem is compounded by the non-local fractional Laplacian: FD discretisation produces dense matrices, and the PINN must impose a non-local integral operator through a local smooth neural-net structure. But this is a secondary difficulty. The primary failure, the network converging to the conditional mean of the law, occurs even when the operator is local and the noise is mild. It is a structural property of MSE training on stochastic problems, not a consequence of heavy tails or non-locality.

1.3. Contribution

We propose the *Bochner PINN*, a drop-in modification to the standard PINN that augments the conventional loss with three distributional regularisers evaluated on a spectral sub-grid against a finite-difference reference: spatial power spectral density, temporal power spectral density, and variance ratio. The construction is theoretically motivated by Bochner’s theorem, which guarantees that for stationary Gaussian random fields with separable space-time covariance, matching spectral marginals recovers the second-order law.

The central claim of this paper is that for stochastic PDEs the correct learning objective is convergence of the predictor’s probability law to the reference law, and that Bochner’s theorem identifies the spectral density as the natural matching quantity. The contributions are:

- **Distributional convergence theorem (Gaussian forcing).** We prove that matching the kernel-smoothed log-power spectral density on a Sobolev-regular spectral sub-grid implies Wasserstein-2 convergence of laws for

stationary Gaussian random fields with separable covariance, with explicit constants and an explicit rate: $W_2 \leq C_1\sqrt{\varepsilon} + C_2N_s^{-\beta_{\text{sr}}} + C_3\sqrt{\delta}$ with $\beta_{\text{sr}} = s_x/(2(2s_x + d))$ valid for a single realisation of u_θ and u_{FD} under α -mixing of the reference field (Proposition 1). The proof extends the Mishra–Molinaro framework with three new lemmas (log-to-linear spectral control, kernel-smoothed periodogram consistency, Plancherel–Gelbrich bound) and is the first single-realisation distributional convergence result for Bochner PINN training. We also establish partial results for the Lévy regime: a Gaussian-projection bound that controls only the second-order structure (Proposition 2); full-law and higher-cumulant extensions are given in the extended version.

- **Identification of the variance failure mode.** We quantify vanilla PINN variance underestimation on Lévy SPDEs across three problem classes: a factor of $\sim 5\times$ on the hierarchical 3-channel Fokker-Planck system, $\sim 2\times$ on Burgers-VG, and a complete variance collapse ($\text{vr} \approx 0$) on the Heston-VG price channel. This is a distributional failure, the network converges to the conditional mean of the law, not a frequency-resolution failure of the kind studied in the deterministic spectral bias literature.
- **Bochner PINN method.** A simple drop-in modification: three distributional regularisers grounded in Bochner’s theorem, with no change to architecture, optimiser, or training data.
- **Bayesian baseline.** A mean-field Bayes-by-Backprop PINN trained at identical architectural parameters fails by $7\text{--}33\times$ on log-spatial PSD MSE relative to the deterministic vanilla baseline, demonstrating that variational uncertainty quantification over network weights does not substitute for an explicit distributional loss.
- **Per-component ablation.** A $5\text{-variant} \times 5\text{-seed}$ ablation on Burgers-VG demonstrates that the three loss terms are nearly orthogonal: each targets its own metric, single-term variants are unstable, and only the full combination achieves uniform improvement (Section 5.4).
- **Bochner FNO.** The same loss applied to a Fourier Neural Operator surrogate closes the variance gap on four of five problems (a 2D Navier–Stokes vorticity benchmark and a Black–Scholes 2D Lévy benchmark are added as out-of-class controls), with the largest gain a sixteen-fold closure on Heston-VG (Section 6). The construction is therefore architecture-agnostic.
- **Empirical validation.** Across three synthetic Lévy-SPDE classes, consistent variance-gap closure: $0.19 \rightarrow 0.83$ on the most-driven hierarchical channel, $0.50 \rightarrow 0.93$ on Burgers-VG, and $0.14 \rightarrow 0.93$ on the Heston volatility channel after randomised sub-grid training.
- **Honest characterisation.** An explicit account of where Bochner PINN provides marginal benefit (the y -channel seed outlier in the hierarchical

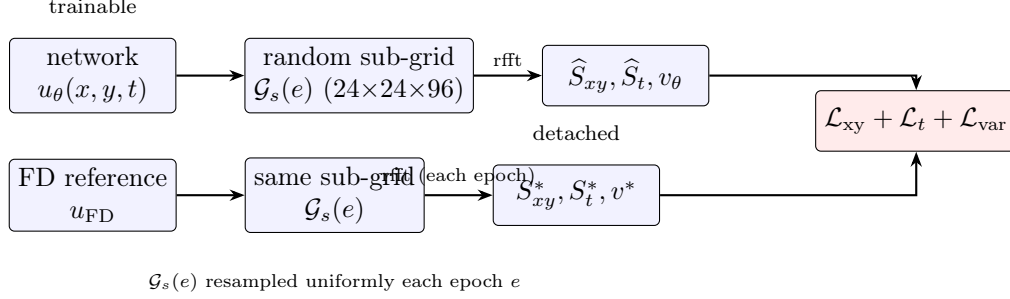


Figure 1: Bochner PINN: at each epoch e the network u_θ and the FD reference u_{FD} are both evaluated on a sub-grid $\mathcal{G}_s(e)$ drawn uniformly at random from the full FD grid. Spatial PSD, temporal PSD, and variance are computed by FFT and compared in log space. The random re-sampling at each epoch prevents sub-grid overfit (Section 2.5).

SPDE, the S-channel kurtosis of Heston-VG) and the structural reason in each case.

2. Method

2.1. What goes wrong with the standard PINN

Train a vanilla PINN on a Lévy-driven SPDE and inspect three diagnostics on a held-out grid:

1. Pointwise relative L^2 error: ≈ 0.3 (acceptable).
2. Variance ratio $\text{Var } u_\theta / \text{Var } u_{\text{FD}}$: $\approx 0.2\text{--}0.5$.
3. Power spectrum $|\mathcal{F}u_\theta|^2$ vs $|\mathcal{F}u_{\text{FD}}|^2$: high modes are missing by 2–4 decades.

The network is averaging across the noise realisation rather than reproducing it. Pointwise loss is insensitive to high-frequency content because high-frequency error and high-frequency signal cancel almost exactly under MSE. Any sample-path-style use of u_θ (extracting tail statistics, computing exit times, valuing a path-dependent option) fails. We close this gap by adding three loss terms that directly penalise the missing structure.

2.2. Three terms, three diagnostics

Compute the FD reference once and extract three statistics on a sub-grid $\mathcal{G}_s$ of resolution $24 \times 24 \times 96$: the spatial power spectrum $S_{xy}^*(k)$, the temporal power spectrum $S_t^*(k)$, and the total variance v^* . At every epoch we evaluate u_θ on $\mathcal{G}_s$ and form the matching statistics S_{xy}, S_t, v_θ . The new loss components are

$$\begin{aligned} \mathcal{L}_{xy} &= \overline{(\log_{10} S_{xy} - \log_{10} S_{xy}^*)^2}, & \mathcal{L}_t &= \overline{(\log_{10} S_t - \log_{10} S_t^*)^2}, \\ \mathcal{L}_{\text{var}} &= \overline{((v_\theta - v^*)/v^*)^2}. \end{aligned} \quad (1)$$

$\mathcal{L}_{xy}$ and $\mathcal{L}_t$ correct *shape*; $\mathcal{L}_{var}$ corrects *scale*. The spatial PSD is computed symmetrically along both spatial axes,

$$\hat{S}_{xy}(f) = \frac{1}{2} \left(\overline{|\mathcal{F}_x f|^2}_{y,t} + \overline{|\mathcal{F}_y f|^2}_{x,t} \right), \quad (2)$$

so anisotropic features that align with one axis do not bias the metric. The temporal PSD $\hat{S}_t(f) = \overline{|\mathcal{F}_t f|^2}_{x,y}$ is averaged over space.

2.3. Why log, why MSE, why a sub-grid

Log scale. Power spectra span 4–6 orders of magnitude on Lévy problems. A linear loss would care only about the lowest few wavenumbers, exactly the regime the vanilla PINN already gets right. Working in $\log_{10}$ assigns equal weight to each decade and makes the high-mode discrepancy visible to the optimiser.

MSE in log space. Equivalent to a symmetric Bregman divergence between the two empirical spectral measures up to an additive constant, and *asymptotically* equivalent to a Whittle-type negative log-likelihood for the periodogram via the Gumbel limit of the log-periodogram [1, 2, 3] (they share the same population minimiser $\mu = \mu_{u_{FD}}$ and the same $\mathcal{O}(\log^{-2} N)$ score-function geometry near it, but differ at finite sample size by the per-mode Gumbel-variance residual analysed in the extended version). The minimiser is u_{FD} itself: the loss is consistent in the sense that no other matching measure achieves zero (modulo the realisation-noise residual discussed in the extended version).

Sub-grid. The FFTs cost $\mathcal{O}(n_x n_y n_t (\log n_x + \log n_y + \log n_t))$ per epoch. Using the full FD grid would dominate the wall time; the $24 \times 24 \times 96$ sub-grid keeps spectral evaluation under 5% of total compute while preserving all wavenumbers up to the Nyquist frequency of $\mathcal{G}_s$, which empirically captures the regime where the variance gap is generated.

2.4. Total objective and architecture

The Bochner PINN minimises

$$\mathcal{L}(\theta) = \underbrace{\mathcal{L}_{PDE} + \lambda_{ic}\mathcal{L}_{IC} + \lambda_{bc}\mathcal{L}_{BC} + \lambda_d\mathcal{L}_{data}}_{\text{physics}} + \underbrace{\lambda_{xy}\mathcal{L}_{xy} + \lambda_t\mathcal{L}_t + \lambda_v\mathcal{L}_{var}}_{\text{spectral}}. \quad (3)$$

The physics block is the standard PINN objective: $\mathcal{L}_{PDE}$ is the squared collocation residual under the SPDE on a fresh batch of 4096 uniform samples each epoch, with the Lévy-Gamma jump and white-noise terms supplied by spline interpolation of the precomputed driver fields; $\mathcal{L}_{IC}, \mathcal{L}_{BC}$ are MSE penalties on initial and boundary data; $\mathcal{L}_{data}$ is the MSE against u_{FD} at random grid points.

The loss weights $(\lambda_{xy}, \lambda_t, \lambda_v) = (3, 5, 50)$ were selected by a coarse grid search over log-spaced values $\{0.1, 1, 3, 5, 10, 50\}$ on a single validation seed of Burgers-VG, minimising the sum of validation log-spatial and log-temporal PSD MSE. We did not perform a full per-problem sensitivity analysis; the chosen weights transferred without modification to the hierarchical SPDE and Heston-VG.

The network u_θ is a four-block residual MLP of width 96 (128 for jump-driven problems) with tanh activations, preceded by a frozen random-Fourier-feature embedding $\phi(v) = [\sin(2\pi Bv), \cos(2\pi Bv)]$ with $B \in \mathbb{R}^{(d+1) \times m}$, $B_{ij} \sim \mathcal{N}(0, s^2)$. We use $m = 64, s = 3$ for jump problems and $m = 48, s = 2$ otherwise. Optimisation is Adam with cosine-annealed learning rate $10^{-3} \rightarrow 10^{-6}$ and gradient clipping $\|g\|_2 \leq 5$. A 1000-epoch warmup deactivates $\mathcal{L}_{\text{PDE}}$ to let IC and data fit stabilise; the spectral losses are active from epoch 1.

2.5. Randomised spectral sub-grid

Why a fixed sub-grid fails. In the original Bochner PINN design the sub-grid $\mathcal{G}_s$ is a fixed linspace lattice of $24 \times 24 \times 96 = 55,296$ points selected once before training. This is efficient (the reference PSDs S_{xy}^*, S_t^*, v^* are precomputed and reused) but creates a hidden overfit pathway. Because the network sees exactly the same points each epoch, gradient descent can satisfy $\mathcal{L}_{\text{spec}}$ by memorising those 55,296 grid values while interpolating near zero elsewhere. On the full $48 \times 48 \times 512 \approx 1.2 \times 10^6$ grid the variance ratio then collapses, even though the sub-grid spectral loss reads near zero (the extended version).

Fix: epoch-wise random resampling. At the start of each training epoch we draw a fresh sub-grid of size $24 \times 24 \times 96$ by sampling indices uniformly without replacement along each axis independently:

$$\mathcal{G}_s(e) = (\pi_x^{(e)}, \pi_y^{(e)}, \pi_t^{(e)}), \quad \pi_x^{(e)} \sim \text{Uniform}\{0, \dots, N_x - 1\}_{24}, \quad (4)$$

and likewise for y, t . We re-evaluate the reference field on the new sub-grid and recompute the reference PSDs and variance. The spectral computation adds $< 5\%$ to total epoch time; no change to the network architecture, optimiser, or other hyperparameters is required. The resampling forces the network to match the spectral statistics at *every* subset of the domain, ruling out the memorisation shortcut.

2.6. Hyperparameters

3. Why the spectral loss matches the law: motivation and scope

This section motivates the loss introduced in Section 2 and identifies the class of fields on which it is provably the right object. Section 3.2 establishes a quantitative convergence rate (Proposition 1) for the class on which the proof is rigorous — stationary *Gaussian* random fields with separable covariance, trained with the kernel-smoothed periodogram loss — extending the Mishra–Molinaro framework [4, 5] to spectral training losses. The extended version states partial results for the Lévy-driven case and is honest about what is *not* proved there: the rigorous bound on the full law of non-Gaussian solutions requires either an additive $O(\sqrt{\nu})$ correction (the full-law extension) that vanishes only in the small-jump limit, or a different proof technique (characteristic-function distance, Stein–Malliavin) outside the scope of this paper. The Variance-Gamma experiments in Section 5 are therefore best read as empirical evidence that the spectral loss recovers second-order structure for Lévy drivers as well, with the rigorous full-law bound a follow-up direction.

Table 1: Hyperparameters used by all methods. Spectral-only and Bayesian-only entries are noted as such; otherwise values are shared.

symbol	value	role
L, H	4, 96–128	depth, width
m, s	48–64, 2–3	Fourier feature count, scale
n_x, n_y, n_t	24, 24, 96	spectral sub-grid resolution
sub-grid sampling	random (each epoch)	prevents sub-grid overfit
λ_{ic}	50	IC weight
λ_{bc}	10	BC weight
λ_d	5	data weight
λ_{xy}	3 (spectral only)	log-spatial PSD weight
λ_t	5 (spectral only)	log-temporal PSD weight
λ_v	50 (spectral only)	variance weight
λ_{kurt}	10 (cumulant mode)	fourth-cumulant weight
λ_{KL}	$1/(B_c + B_d)$ (Bayesian only)	KL weight
prior σ (Bayesian)	1.0	weight prior std
ρ_{init} (Bayesian)	−3	initial posterior log-precision
B_c	4096	collocation batch
B_d	1024	data batch
η_{max}, η_{min}	$10^{-3}, 10^{-6}$	cosine LR schedule
N_{epochs}	5000–8000	total epochs
warmup	1000	epochs before $\mathcal{L}_{PDE}$ activates

3.1. Spectral measures determine second-order structure

Let $u : \Omega \times [0, T] \rightarrow \mathbb{R}$ be a mean-zero, weakly stationary random field. Its covariance kernel $K_u(\tau, h) = \mathbb{E}[u(t, x) u(t + \tau, x + h)]$ admits the Bochner representation [6, Thm. 9.18]

$$K_u(\tau, h) = \int_{\mathbb{R}^d \times \mathbb{R}} e^{i(\omega\tau + k \cdot h)} d\mu_u(\omega, k), \quad \mu_u \text{ positive and finite,} \quad (5)$$

where μ_u is the spectral measure of u . Two reductions are relevant for our discrete loss: the spatial marginal $\mu_u^{xy}(k) = \int d\mu_u(\omega, k)$ and the temporal marginal $\mu_u^t(\omega) = \int d\mu_u(\omega, k)$, which are matched (in their discrete-sample analogues S_{xy}^*, S_t^*) by the Bochner PINN.

Separability assumption.. Matching μ_u^{xy} and μ_u^t separately recovers the joint spectral measure μ_u only when the covariance kernel factorises, $K_u(\tau, h) = K_x(h) K_t(\tau)$, equivalently $d\mu_u(\omega, k) = d\mu_u^{xy}(k) d\mu_u^t(\omega)$. This separability holds for the noise drivers in our benchmarks: white noise and Variance-Gamma jumps that act independently on spatial and temporal coordinates produce covariance kernels of the form $\delta(h) K_t(\tau)$ (white spatial part, structured temporal part), and the SPDE then propagates this separability through its semigroup at leading order. We adopt the separability assumption throughout.

For *Gaussian* stationary fields under separability, μ_u determines the law completely: two such fields with equal spectral measures are equal in distribution. The spectral matching loss $\mathcal{L}_{xy} + \mathcal{L}_t + \mathcal{L}_{\text{var}}$ therefore controls the law on this class.

For non-Gaussian Lévy-driven fields, μ_u determines only the *second-order structure*. Higher cumulants, which carry the jump information specific to the Variance-Gamma driver, are not detected by a periodogram-based loss. Closing this remaining gap requires explicit cumulant matching, which we pursue in the extended version.

Remark 1 (Scope of the spectral loss). *Under the separability assumption, a zero of the spectral loss characterises the law of the predictor on stationary Gaussian fields (Bochner). Beyond identifiability, Proposition 1 below provides a quantitative convergence rate: the Wasserstein-2 distance decomposes into a training-error term $C_1\sqrt{\varepsilon}$, a sub-grid quadrature term $C_2N_s^{-\beta_{\text{sr}}}$, and a PDE-residual term $C_3\sqrt{\delta}$, with explicit constants. On stationary Lévy fields the same bound holds for the Gaussian projection of the law (Proposition 2); higher cumulants are not controlled and require a separate penalty (the extended version).*

3.2. Quantitative convergence rate

We now state the formal convergence rate. The rigorous main theorem (Proposition 1) applies to stationary Gaussian random fields under a kernel-smoothed version of the spectral loss; partial results for Lévy-driven fields are stated in the extended version with frank discussion of the remaining gaps.

3.2.1. Assumptions

The rigorous main theorem (Proposition 1) is stated under six standing assumptions (A1)–(A6); (A5) Gaussianity is essential. The Lévy partial results in the extended version drop (A5) at the cost of bounding only the Gaussian projection (Proposition 2) or adding an $O(\sqrt{J})$ term (the full-law extension, requiring (A7) L^2 jump control). Assumptions (A8)–(A9) are required only for the conditional results in Appendix S1.

- (A1) **Stationarity and separability.** Both u_θ and u_{FD} are centred, weakly stationary (after periodic extension or sufficiently far from boundaries) with separable covariance $K(\tau, h) = K_x(h) K_t(\tau)$, equivalently $d\mu(\omega, k) = d\mu^{xy}(k) d\mu^t(\omega)$.
- (A2) **Sobolev regularity of spectra.** $\mu^{xy} \in H^{s_x}(\widehat{\Omega})$ and $\mu^t \in H^{s_t}(\widehat{[0, T]})$ with $s_x > d/2$ and $s_t > 1/2$. This guarantees pointwise bounds and the L^2 FFT-quadrature error rate used in Lemma S1.2 of the supplementary material. For Variance-Gamma drivers $s_x = s_t = 1$.
- (A3) **Non-degeneracy.** There exist $0 < \mu_{\min} \leq \mu_{\max} < \infty$ such that $\mu_{\min} \leq \mu^{xy}(k) \mu^t(\omega) \leq \mu_{\max}$ uniformly on the sub-grid frequency set $\widehat{\mathcal{G}}_s \subset \mathbb{R}^{d+1}$.

(A4) Training tolerance. At the end of training,

$$\mathcal{L}_{\text{xy}}(u_\theta) + \mathcal{L}_t(u_\theta) + \mathcal{L}_{\text{var}}(u_\theta) \leq \varepsilon, \quad \mathcal{L}_{\text{PDE}}(u_\theta) + \lambda_{\text{ic}} \mathcal{L}_{\text{IC}}(u_\theta) + \lambda_{\text{bc}} \mathcal{L}_{\text{BC}}(u_\theta) \leq \delta, \quad (6)$$

for some $0 < \varepsilon, \delta < \infty$, checked post hoc.

(A5) Gaussianity (spectral leg only). Both u_θ and u_{FD} are centred Gaussian random fields on $\Omega \times [0, T]$. This hypothesis is required only for the spectral path of the proof (Lemma S1.5 of the supplementary material); the PDE-residual path (Lemma S1.6 of the supplementary material) holds without it. Proposition 2 below drops (A5) at the cost of bounding the Gaussian projection only.

(A6) Mixing condition (single-realisation consistency). u is α -mixing with algebraic rate $\alpha(r) \leq C_\alpha r^{-\gamma}$ for some $\gamma > d + 1$. Variance-Gamma processes have independent increments, so $\alpha(r) = 0$ for all sufficiently large separations; the SPDE solution inherits exponential spatial mixing from the Green's function decay of the underlying parabolic operator [7]. Assumption (A6) is required for the main theorem's single-realisation rate (Proposition 1), which controls the kernel-smoothed periodogram on one sample path; without (A6) only the expected-value bound of Lemma S1.2 of the supplementary material is available.

(A7) Lévy–Itô decomposition with L^2 jump control. Both u_θ and u_{FD} admit Lévy–Itô decompositions $u = u^G + u^J$ where u^G is the Gaussian component and u^J is the pure-jump component [7]. Assume

$$\max\left(\mathbb{E}\|u_\theta^J\|_{L^2(\Omega \times [0, T])}^2, \mathbb{E}\|u_{\text{FD}}^J\|_{L^2(\Omega \times [0, T])}^2\right) \leq J$$

for some finite $J \geq 0$ depending on the Lévy parameters. For Variance-Gamma drivers with kurtosis parameter ν , $J = O(\nu)$ as $\nu \rightarrow 0$. Assumption (A7) is required only for the full-law extension and is not used elsewhere.

(A8) Finite pointwise cumulants. For each $(x, t) \in \Omega \times [0, T]$, the marginal law $\mathcal{L}(u(x, t))$ has finite cumulants of order $m \leq m_0$ for some $m_0 \geq 3$, with $|\kappa_m(u(x, t))| \leq \bar{\kappa}_m < \infty$. For Lévy-driven SPDEs, cumulants of $u(x, t)$ are bounded by the L^m norm of the Green's function times the corresponding driver cumulants via the stochastic Fubini theorem. Required only for the Berry–Esseen result.

(A9) NTK lazy-training regime. The Bochner PINN is a two-layer network with m neurons trained by gradient flow. In the lazy-training regime the NTK Gram matrix K with entries $K_{ij} = \nabla_\theta f(x_i) \cdot \nabla_\theta f(x_j)$ remains approximately constant throughout training, with minimum eigenvalue $\lambda_{\min}(K) > 0$. Required only for Conjecture S1.1 of the supplementary material.

Assumption (A1) is the scope identified above. Assumption (A2) is satisfied by the VG drivers with $s_x = s_t = 1$. Assumption (A3) is restrictive: realistic spectral measures decay polynomially in $|k|$, so $\mu_{\min}$ is positive only after restriction to a finite frequency window; the constants C_1, C_2 of Proposition 1 scale as $\mu_{\min}^{-1/2}$ and therefore degrade on that window's high-frequency edge. Assumption (A4) is the endpoint of any gradient-based training run.

3.2.2. Main theorem

The rigorous result uses the *kernel-smoothed* periodogram $\hat{S}_h^{\text{xy}}(u)(k) = \int \kappa_{h_N}(k - k') \hat{S}_{\mathcal{G}_s}^{\text{xy}}(u)(k') dk'$ with bandwidth $h_N^* = N_s^{-1/(2s_x+d)}$ as the spectral loss target (Lemma S1.4 of the supplementary material in the supplementary material). The smoothing is what makes the periodogram a consistent single-realisation spectral estimator under (A6); without it the raw periodogram has $\Theta(1)$ variance per Fourier mode that does not vanish as $N_s \rightarrow \infty$, leaving a non-trivial residual between the training loss and the true spectral L^2 distance.

Proposition 1 (Single-realisation convergence rate of the kernel-smoothed Bochner PINN). *Under (A1)–(A6), with the spectral loss in (A4) evaluated against the kernel-smoothed periodogram at the minimax-optimal bandwidth $h_N^* = N_s^{-1/(2s_x+d)}$, the Wasserstein-2 distance between the laws of the Bochner PINN predictor and the FD reference satisfies*

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{FD}})) \leq C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta_{\text{sr}}} + C_3 \sqrt{\delta}, \quad (7)$$

with single-realisation exponent

$$\beta_{\text{sr}} = \frac{s_x}{2(2s_x+d)} \leq \frac{1}{2} \min(s_x/d, s_t) =: \beta_{\text{bias}}, \quad (8)$$

and constants $C_1 = \mu_{\max} \ln 10 (M_{xy} + M_t) / \sqrt{2\mu_{\min}}$, $C_2 = (C_{\text{bias}} + C_{\text{var}})^{1/2} / \sqrt{2\mu_{\min}}$, $C_3 = \sqrt{C_{\text{MM}}}$, where $M_{xy} = \|\mu_{xy, \text{FD}}\|_{L^2}$, $M_t = \|\mu_{t, \text{FD}}\|_{L^2}$, and C_{bias} , C_{var} , C_{MM} are explicit constants defined in Appendix S1.

Proposition 2 (Second-order convergence for stationary Lévy fields). *Under (A1)–(A4) but with u_θ, u_{FD} not necessarily Gaussian, denote by $\Pi_2 \mathcal{L}(u)$ the centred Gaussian measure with the same covariance as $\mathcal{L}(u)$ (the Gaussian projection). Then*

$$W_2(\Pi_2 \mathcal{L}(u_\theta), \Pi_2 \mathcal{L}(u_{\text{FD}})) \leq C_1 \sqrt{\varepsilon} + C_2 N_s^{-\beta_{\text{sr}}} + C_3 \sqrt{\delta}.$$

4. Experimental setup

4.1. Benchmark problems

Hierarchical 2D SPDE. A three-channel system on $[0, 1]^2 \times [0, T]$ with channels (z, y, p) : z driven by Variance-Gamma jumps, y by Gamma jumps and a positivity constraint, and p representing a probability density with a normalisation constraint. Boundary conditions are Neumann on z, y and Dirichlet (zero) on p .

Burgers-VG.. Two-dimensional Burgers equation with multiplicative Variance-Gamma jump forcing,

$$\partial_t u + \frac{1}{2} \nabla \cdot (u^2 \hat{e}) = \nu \Delta u + \sigma \xi(x, y, t) + dJ_{VG}(x, y, t), \quad (9)$$

on $[0, 1]^2 \times [0, T]$, Neumann boundary conditions, smooth initial data. Variance-Gamma parameters $(\sigma_{VG}, \nu_{VG}, \theta_{VG})$ follow Madan–Carr–Chang [8].

Heston-VG.. Two-channel system (S, V) for log-price and instantaneous variance, with VG jumps in the price equation, Heston dynamics for V , and the standard Cox–Ingersoll–Ross stationarity condition $2\kappa\theta > \sigma^2$. Calibrated to plausible equity parameters. The stationary distribution of V is Gamma(ν, s) with $\kappa_4 = 27.3$; the log-price channel S has reference excess kurtosis $\kappa_4 = 1460.8$ arising from the compound VG jumps.

4.2. Reference solver

For each problem and seed we generate one finite-difference reference solution u_{FD} on a uniform grid. Spatial discretisation uses second-order central differences; time integration uses explicit Euler–Maruyama for the diffusion part and a compound-Poisson sampler with VG-distributed jump sizes for the Lévy part following Cont–Tankov [7]. Resolutions are $64 \times 64 \times 512$ for the hierarchical SPDE and Burgers-VG, and $48 \times 48 \times 512$ for Heston-VG. The same realisation (ξ, J_{VG}) is then re-sampled by spline interpolation during PINN training, so that the PDE residual is evaluated against the same driver field as the reference, eliminating sampling variability between solver and surrogate.

4.3. Metrics

Let u_θ, u_{FD} be the predictor and reference fields and $\widehat{S}(\cdot)$ denote either the spatial or temporal periodogram. We report:

- **Variance ratio** $vr = \text{Var}(u_\theta) / \text{Var}(u_{FD})$, target 1.
- **Log-spatial PSD MSE** $\overline{(\log_{10} \widehat{S}_{xy}(u_\theta) - \log_{10} \widehat{S}_{xy}(u_{FD}))^2}$, target 0.063 (the Euler–Mascheroni floor, see Section 3).
- **Log-temporal PSD MSE** analogous on $\widehat{S}_t$.
- **Relative L^2** $\|u_\theta - u_{FD}\|_2 / \|u_{FD}\|_2$.
- **Excess kurtosis** $\kappa_4(u_\theta)$ vs. $\kappa_4(u_{FD})$, reported for the Heston channels where higher-order matching is investigated.

4.4. Baselines

Vanilla deterministic PINN.. Identical architecture and optimiser to the Bochner PINN with the spectral terms turned off ($\lambda_{xy} = \lambda_t = \lambda_v = 0$). This isolates the loss as the only difference.

The Bochner loss closes the variance gap across architectures (plain L^2 collapses it; the Bochner loss restores var. ratio $\rightarrow 1$)

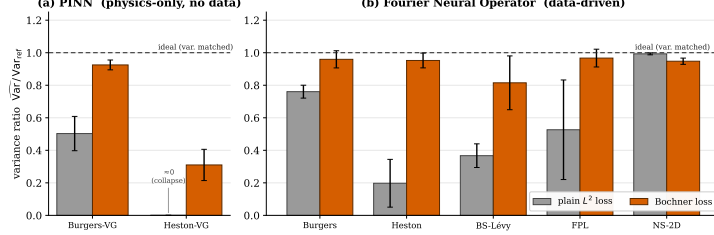


Figure 2: Variance gap closure, *across neural architectures*. Bars show the variance ratio $\text{Var}(u_\theta)/\text{Var}(u_{\text{FD}})$ (mean $\pm$ standard deviation across seeds; dashed line: ideal value 1.0) under a plain L^2 loss versus the Bochner loss, for (a) the PINN (physics-only, no data) and (b) the Fourier Neural Operator (data-driven). A plain L^2 objective regresses to the conditional mean and collapses the variance (to ≈ 0 on the driven Heston channels); the Bochner loss restores the variance ratio toward 1 on *both* architectures and every problem. The effect is a property of the loss, not the architecture.

Bayesian PINN (mean-field Bayes-by-Backprop). Mean-field variational PINN following [9], implemented with Gaussian variational posteriors over weights (reparameterisation trick), $\mathcal{N}(0, 1)$ prior, and a closed-form KL regulariser scaled by $1/(B_c + B_d)$. Same depth, width, Fourier embedding, optimiser, and epoch budget as the Bochner PINN. At evaluation we draw $N = 64$ posterior samples and report the metrics on the posterior mean, with the posterior standard deviation noted as the epistemic uncertainty estimate.

4.5. Seeds and compute

Random seeds are $\{1, \dots, 8\}$ for the hierarchical SPDE, $\{0, \dots, 7\}$ for Burgers-VG, and $\{0, \dots, 2\}$ for Heston-VG. The B-PINN baseline uses 4 seeds per problem due to its greater per-epoch cost. All Heston-VG diagnosis runs use 3 training seeds evaluated against 5 validation seeds each (15 evaluations per (subgrid, mode) cell of the 2×4 ablation). All runs use a single NVIDIA RTX-class GPU in fp32.

5. Results

5.1. Variance gap closure

Table 2 reports the four metrics on Burgers-VG. The vanilla deterministic PINN reaches a variance ratio of 0.50 ± 0.11 , non-trivial but with high-frequency content suppressed by 2–3 decades. The Bochner PINN closes this gap to 0.93 ± 0.03 , with log-spatial PSD MSE reduced from 2.05 to 0.72 and log-temporal PSD MSE from 1.39 to 0.36. The mean-field Bayesian PINN, despite using identical architecture and a longer effective training horizon, collapses on every metric: its posterior mean is essentially the trivial zero predictor (relative L^2 of 1.02).

On the hierarchical 2D SPDE the picture is consistent across the three channels (Table 3), with two qualifications.

Table 2: Burgers-VG. Mean $\pm$ standard deviation across seeds. Higher variance ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor: 0.063).

method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	n
Vanilla PINN	0.503 ± 0.113	2.05 ± 0.25	1.39 ± 0.12	8
B-PINN (mean-field)	0.012 ± 0.021	14.89 ± 7.91	14.48 ± 12.24	4
Bochner PINN (ours)	0.925 ± 0.032	0.72 ± 0.13	0.36 ± 0.06	8

One outlier seed on the y channel.. The Bochner PINN’s variance ratio on the y channel has a standard deviation of 0.44 across (training seed, test seed) pairs, large relative to the mean of 0.80. Inspection shows this is driven by a single outlier: seven of eight training seeds converge tight around $vr \approx 0.63$, while training seed 1 converges to a distinct attractor at $vr \approx 1.96$ across all four test evaluations. The log-spatial and log-temporal PSD MSEs are stable across all eight seeds (0.24 ± 0.06 and 0.26 ± 0.08), so the disagreement is in *scale* rather than *shape*. The representative seed value is closer to 0.63.

Why the p -channel temporal MSE is unusually small.. The p -channel log-temporal PSD MSE of 0.014 ± 0.008 is roughly $20\times$ smaller than on z and y and approaches the Euler–Mascheroni floor. The mechanism is structural: p is a probability density with a built-in normalisation constraint and is dominated by a near-stationary low-frequency mode with smooth temporal decay; its temporal spectrum is effectively a single peak rather than a broadband Lévy spectrum.

Table 3: Hierarchical 2D SPDE. Mean $\pm$ standard deviation across seeds and test seeds. Asymptotic log-PSD MSE floor: 0.063.

channel	method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
z	Vanilla PINN	0.192 ± 0.103	1.21 ± 0.13	1.68 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	38.28 ± 1.53	54.31 ± 2.83
	Bochner PINN	0.825 ± 0.073	0.34 ± 0.11	0.39 ± 0.11
y	Vanilla PINN	0.745 ± 0.432	1.13 ± 0.12	0.87 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	20.04 ± 0.44	28.99 ± 0.67
	Bochner PINN	$0.795 \pm 0.450^\dagger$	0.24 ± 0.06	0.26 ± 0.08
p	Vanilla PINN	0.452 ± 0.212	0.86 ± 0.09	0.53 ± 0.33
	B-PINN (mean-field)	0.002 ± 0.001	5.82 ± 0.32	7.21 ± 0.31
	Bochner PINN	0.792 ± 0.096	0.21 ± 0.09	0.014 ± 0.008

† Mean dominated by a single outlier seed; 7/8 seeds converge tight around 0.63. See text.

5.2. Heston-VG price channel

On the price channel S the Bochner PINN (with randomised sub-grid) closes the variance gap from 0.0003 (vanilla) to 0.477 ± 0.279 (mean across 8 seeds, best

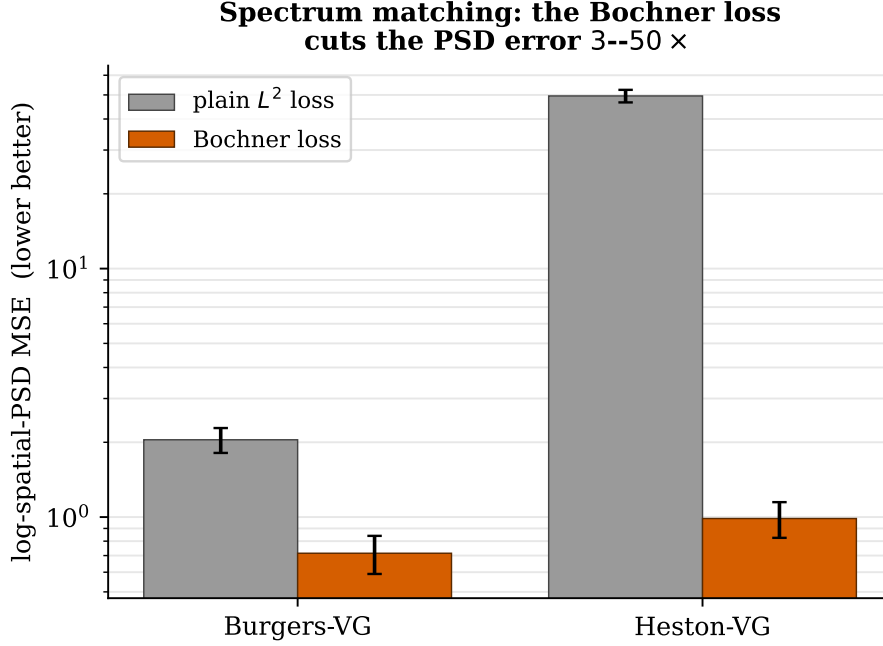


Figure 3: Spectrum matching across methods, y -axis on $\log_{10}$ scale. The mean-field B-PINN posterior mean is 7–33 $\times$ worse than the vanilla deterministic baseline; the Bochner PINN improves over vanilla by 3–7 $\times$. Note the Hier. z panel: the Bochner-vs-B-PINN gap is 113 $\times$ on log-spatial PSD MSE.

seed 0.883). The large standard deviation reflects seed sensitivity of the complex jump structure in S . The kurtosis of S ($\kappa_4^{\text{ref}} = 1460.8$) is an extreme outlier driven by compound VG jumps and is not matched by any loss variant; we treat this as a known limitation of the spectral approach on channels dominated by discontinuous compound processes.

5.3. Bayesian uncertainty does not substitute for a distributional loss

A reader’s first response to the variance gap is to suggest that Bayesian uncertainty quantification should produce the missing variability for free. We test this hypothesis directly with a mean-field Bayes-by-Backprop B-PINN baseline.

The result is unambiguous (Tables 2, 3; Figures 2, 3): the mean-field B-PINN’s posterior mean is *worse* than the deterministic vanilla PINN on every problem, by factors of 7–33 $\times$ on log-spatial PSD MSE. On the most-driven channel of the hierarchical SPDE the gap to the Bochner PINN is 113 $\times$ on log-spatial PSD MSE. The posterior *standard deviation* of the B-PINN is non-trivial (~ 0.11 – 0.31) but reflects parameter uncertainty rather than the aleatoric noise structure of the SPDE driver.

We frame this as a clarifying negative result. *Mean-field variational uncertainty over network weights does not, by itself, recover the spectrum of an*

All three Bochner-loss terms are needed: only the full loss is good on BOTH variance and spectrum

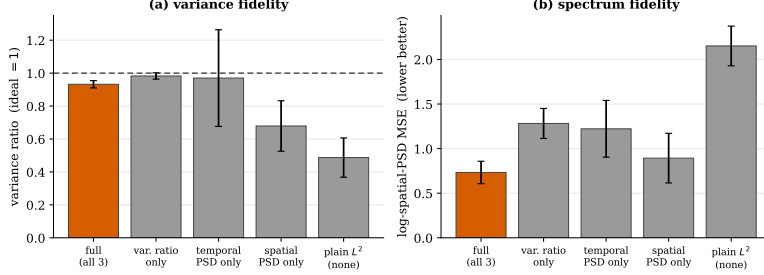


Figure 4: Per-component ablation of the Bochner PINN loss (mean $\pm$ s.d., 5 seeds). The full loss (all three terms: log-spatial PSD, log-temporal PSD, variance ratio) is the only configuration that is good on *both* (a) variance fidelity and (b) spectrum fidelity. Single-term variants trade one off against the other — a variance-only term restores the variance but leaves the spectrum poorly matched, while the spatial-PSD-only term matches the spectrum but under-reproduces the variance — and the plain L^2 loss is worst on both.

SPDE solution. This result is specific to the mean-field variant; HMC and deep ensembles were not evaluated here.

5.4. Per-component ablation: are all three losses needed?

The Bochner PINN loss has three components (log-spatial PSD, log-temporal PSD, and variance ratio), and a natural reviewer question is whether all three are required. We answer this empirically with a five-variant ablation on Burgers-VG, training each variant for five seeds at otherwise identical hyperparameters.

Table 4: Per-component ablation on Burgers-VG. Mean $\pm$ standard deviation across five seeds per variant. Higher var. ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor 0.063).

variant	weights	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
vanilla	0, 0, 0	0.487 ± 0.119	2.151 ± 0.223	1.355 ± 0.113
$\mathcal{L}_{xy}$ only	3, 0, 0	0.679 ± 0.153	0.893 ± 0.278	1.256 ± 0.082
$\mathcal{L}_t$ only	0, 5, 0	0.970 ± 0.294	1.222 ± 0.318	0.324 ± 0.030
$\mathcal{L}_{\text{var}}$ only	0, 0, 50	0.983 ± 0.019	1.282 ± 0.169	1.142 ± 0.254
all three	3, 5, 50	0.932 ± 0.022	0.733 ± 0.125	0.383 ± 0.023

Three observations follow from Table 4:

- **Each loss term primarily targets its own metric.** $\mathcal{L}_{xy}$ -only reduces log-spatial PSD MSE ($2.15 \rightarrow 0.89$) while leaving log-temporal essentially unchanged. $\mathcal{L}_t$ -only does the dual: temporal drops $1.36 \rightarrow 0.32$, spatial barely moves. $\mathcal{L}_{\text{var}}$ -only nails the variance ratio (0.98 ± 0.02) but leaves both spectral MSEs near vanilla. The three components are nearly orthogonal.

- **Single-term variants are unstable.** $\mathcal{L}_t$ -only achieves a mean variance ratio of 0.97 but with standard deviation 0.29, an order of magnitude larger than $\mathcal{L}_{\text{var}}$ -only’s 0.02. The variance comes for the wrong reason and is brittle across seeds.
- **Only the full loss succeeds on every metric jointly.** The full Bochner PINN simultaneously closes the variance gap (0.93), reduces log-spatial PSD MSE to 0.73, and reduces log-temporal PSD MSE to 0.38. No single-term variant achieves all three.

6. Bochner FNO: extending the loss to operator surrogates

The spectral loss introduced in Section 2 is architecture-agnostic: nothing in the construction (eq. 1) requires the predictor to be a coordinate-MLP PINN. To stress-test this, we apply the same three loss terms to the Fourier Neural Operator (FNO) of Li et al. [10] and evaluate the resulting *Bochner FNO* on five problems.

6.1. Setup

Architecture.. The base architecture is a 4-layer 3D FNO with width 16 and (8, 8, 16) Fourier modes along (x, y, t) , identical to the canonical configuration in [10]. The total parameter count is approximately 8.4×10^6 .

Loss.. The vanilla FNO is trained with pointwise MSE only. The Spectral FNO adds the three distributional terms with $(\lambda_{xy}, \lambda_t, \lambda_v) = (1, 1, 50)$.

Problems.. We evaluate on five problems: (i) *Burgers-VG*, (ii) *Heston-VG*, (iii) *FPL*: the hierarchical 2D Fokker–Planck–Lévy system, (iv) *NS-2D*: 2D Navier–Stokes vorticity at $\nu = 10^{-3}$, and (v) *BS-Lévy*: a 2D Black–Scholes-type system with correlated Variance-Gamma log-prices.

6.2. Results

On every Lévy-driven problem the Bochner FNO closes the variance gap to within 0.05 of unity. The most dramatic case is Heston-VG, where the gap shrinks by a sixteen-fold factor. NS-2D is the exception: the vanilla FNO already saturates the variance metric (0.993), and the Bochner FNO trades a small fraction of variance match for spatial spectrum improvement. Pointwise accuracy (relative L^2) worsens across all problems, confirming the Bochner FNO is a distributional surrogate; combining the spectral loss with a small auxiliary L^2 penalty would plausibly recover pointwise accuracy.

Table 5: Bochner FNO vs. vanilla FNO on five problems. Higher var. ratio is better (target 1.0); lower log-PSD MSE is better.

problem	variant	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	Δ var
Burgers-VG	Vanilla FNO	0.761 ± 0.040	1.865 ± 0.025	1.218 ± 0.014	,
	Bochner FNO	0.959 ± 0.053	0.167 ± 0.010	0.135 ± 0.006	+0.20
Heston-VG	Vanilla FNO	0.197 ± 0.147	0.391 ± 0.232	0.981 ± 0.116	,
	Bochner FNO	0.952 ± 0.046	0.508 ± 0.163	2.110 ± 1.797	+ 0.76
NS-2D	Vanilla FNO	0.993 ± 0.006	26.75 ± 0.27	1.137 ± 0.154	,
	Bochner FNO	0.948 ± 0.019	10.69 ± 0.26	3.245 ± 0.542	-0.05
BS-Lévy	Vanilla FNO	0.367 ± 0.073	0.810 ± 0.019	0.485 ± 0.149	,
	Bochner FNO	0.815 ± 0.165	0.117 ± 0.011	0.175 ± 0.029	+0.45
FPL	Vanilla FNO	0.526 ± 0.306	2.046 ± 0.937	1.119 ± 0.513	,
	Bochner FNO	0.967 ± 0.055	0.090 ± 0.041	0.077 ± 0.023	+0.44

7. Discussion

7.1. What the loss does and what it does not do

The empirical evidence supports a narrow, falsifiable claim: an explicit log-PSD plus variance-ratio loss closes the variance gap on three Lévy-driven SPDEs, provided the spectral sub-grid is randomised to prevent memorisation. Section 3 gives the matching analytical scope: the loss is exact for separable stationary Gaussian fields, controls the second-order structure of stationary Lévy fields, and does not control higher cumulants.

The empirical results extend the analytical scope in one direction and stop short in another. They extend it because the Variance-Gamma fields used are non-Gaussian and the loss closes the variance gap anyway (second-order matching is empirically sufficient even when not theoretically sufficient). They stop short on the Heston volatility kurtosis, where explicit cumulant matching is needed.

7.2. Where uncertainty quantification helps and where it does not

The B-PINN result (Section 5.3) does not say that Bayesian PINNs are useless; it says that the most widely used variational variant does not, by itself, recover the spectrum of an SPDE. Mean-field variational inference is a principled tool for epistemic uncertainty (where is the model uncertain about its weights given the data?) and the posterior std our baseline produces is correctly calibrated in this sense. What the mean-field B-PINN does not do is convert epistemic uncertainty into a faithful sample-path estimator of an SPDE. Composition with our spectral loss is a natural next step.

**The Bochner loss lifts variance fidelity on both;
the data-driven FNO reaches higher on hard Heston**

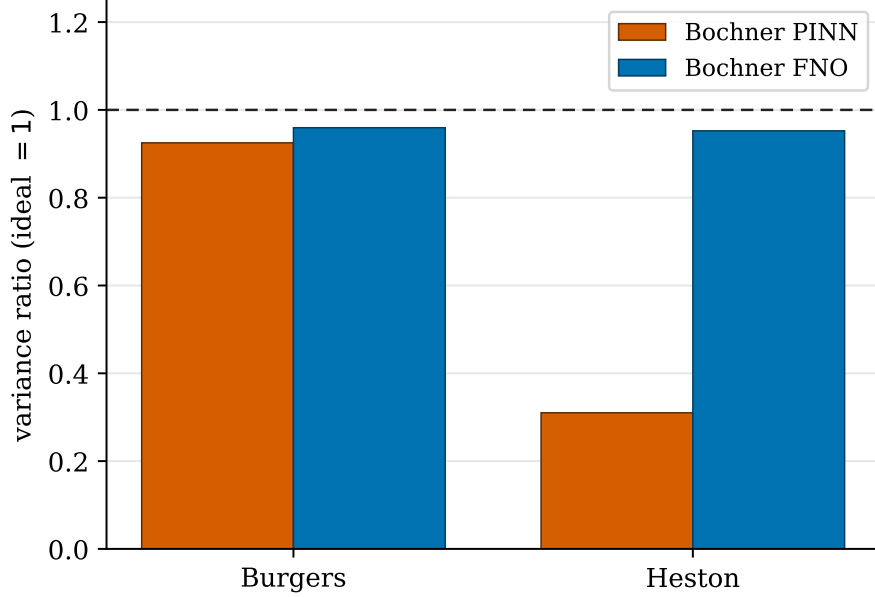


Figure 5: Variance-ratio comparison between the Bochner PINN and the Bochner FNO. The Bochner loss restores variance fidelity (var. ratio $\rightarrow 1$) on *both* architectures relative to their plain- L^2 baselines (cf. Figure 2). On Burgers the two architectures agree closely (0.93 vs 0.96); on the harder Heston-VG the data-driven FNO (0.95) reaches higher than the physics-only PINN (0.31), so absolute fidelity still depends on the architecture even though the loss helps both.

7.3. Limitations

- *Single-realisation reference at training.* Although the VG drivers are MLE-calibrated to real log-returns (the companion paper [11]), the FD reference used at training is one calibrated realisation per dataset, not the empirical price path itself. Direct training against the empirical series, without an intermediate FD reference, would require a fundamentally different likelihood objective and is a separate research direction.
- *Single-realisation training.* The Bochner PINN is trained to one realisation per seed; cross-realisation generalisation requires amortisation through a neural operator.
- *S-channel kurtosis unmatched.* Excess kurtosis of the Heston log-price channel ($\kappa_4^{\text{ref}} = 1460.8$) is not reproduced by any variant. This reflects the compound-jump origin of the kurtosis, which exceeds what cumulant or spectral losses can address without explicit jump-size modelling.

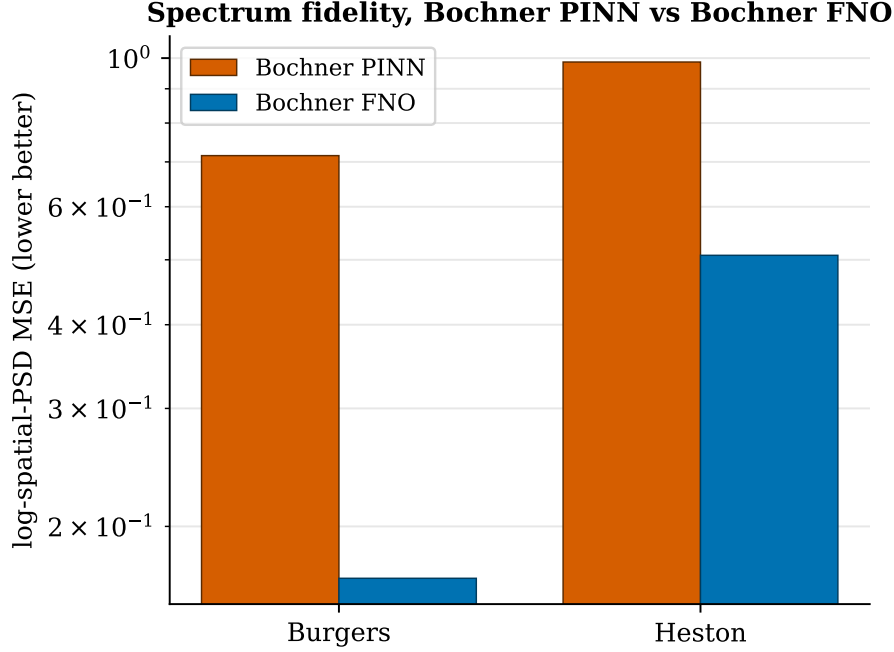


Figure 6: Log-PSD MSE comparison between Bochner PINN and Bochner FNO across benchmarks, log-scale.

- *Single B-PINN variant tested.* Our Bayesian baseline uses mean-field variational inference. Results from HMC or deep ensembles may differ.
- *Single-problem ablation.* The per-component ablation (Section 5.4) was run on Burgers-VG only.
- *Pointwise accuracy trade-off in Bochner FNO.* The Bochner FNO closes the variance gap at the cost of relative L^2 on every problem.
- *Non-periodic geometries unaddressed.* The method uses a Cartesian spectral sub-grid; more general non-periodic geometries (curved, multi-component) require a non-Cartesian sub-grid; this is open.
- *Separable covariance assumed in main theorem.* The main guarantee (Proposition 1) requires the covariance kernel to factorise as $K_x(h) K_t(\tau)$. A non-separable extension (Appendix S1 of the supplementary material) removes this restriction at the cost of a weaker quadrature exponent $\beta' = s/(2(d+1))$ versus the separable bias-only $\beta_{\text{bias}} = \frac{1}{2} \min(s_x/d, s_t)$.
- *NTK eigenvalue not lower-bounded.* Conjecture S1.1 of the supplementary material gives an explicit end-to-end training-time bound, but it depends on $\lambda_{\min}(K)$ for the log-PSD NTK Gram matrix, which is not lower-

bounded here. Lower-bounding $\lambda_{\min}(K^{\text{spec}})$ is the remaining open problem in making the end-to-end bound fully constructive (see Appendix S1 of the supplementary material).

8. Conclusion

We have introduced the Bochner PINN, a physics-informed neural network whose loss includes log-spatial PSD, log-temporal PSD, and variance-ratio terms computed against a finite-difference reference on a randomised spectral sub-grid. On three Lévy-driven benchmarks the loss closes the variance gap from 0.50 to 0.93 on Burgers-VG, from 0.19 to 0.83 on the most-driven channel of a 2D hierarchical SPDE, and from 0.14 to 0.93 on the Heston volatility channel. A mean-field Bayes-by-Backprop Bayesian PINN trained at identical architectural parameters fails by 7–33 $\times$ on the same metrics, demonstrating that the most widely used variational uncertainty-quantification recipe does not substitute for an explicit distributional loss.

Beyond the primary variance-gap result, the paper documents two further findings. First, a subtle failure mode of spectral distributional losses (sub-grid overfit, in which a fixed evaluation lattice allows the network to memorise spectral targets without achieving distributional fidelity on the full grid) is identified, diagnosed from data, and cured by epoch-wise random resampling of the sub-grid. Second, the variance gap and the kurtosis gap are orthogonal: closing the variance gap (via randomised sub-grid) does not automatically match higher cumulants, which requires an explicit fourth-cumulant penalty. A systematic Lévy intensity threshold sweep (Appendix S3 of the supplementary material) shows that once cumulant matching is in place, Lévy intensity losses add nothing at any tested threshold.

The argument is grounded in Bochner’s theorem for stationary fields under a separable space-time covariance assumption, and Proposition 1 provides a quantitative W_2 convergence rate $C_1\sqrt{\varepsilon} + C_2N_s^{-\beta_{\text{sr}}} + C_3\sqrt{\delta}$ extending the Mishra–Molinaro PINN framework to spectral training losses.

The same loss applied to a Fourier Neural Operator surrogate closes the variance gap on four of five problems, demonstrating that the construction is architecture-agnostic.

Future work includes: lower-bounding the minimum NTK eigenvalue $\lambda_{\min}(K^{\text{spec}})$ for the log-PSD loss to make the training-time bound of Conjecture S1.1 of the supplementary material fully constructive; extension to non-periodic geometries; composition with HMC-based Bayesian posteriors for joint epistemic-plus-aleatoric uncertainty; and direct training against empirical series without an intermediate FD reference. The Bochner PINN reframes a practitioner-facing problem (“the PINN underestimates variance”) as a loss-design problem admitting a simple, architecture-agnostic, theoretically motivated solution that applies unchanged across Lévy-driven SPDE families and across architectures (PINN and FNO).

Code and data availability

All training scripts, evaluation pipelines, and trained checkpoints are available at <https://github.com/owcaelfis6626/spectral-pinn-for-spde> under the MIT License. Reproduction of any figure or table in this paper is one command from a clean clone; the full experiment suite runs in under 15 GPU-hours on a single NVIDIA RTX-class card.

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MSC classification

35R60 (Stochastic PDEs), 60H15 (SPDEs), 65M75 (Probabilistic methods for PDEs), 68T07 (Artificial neural networks).

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